

Tiago da Silva

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Education

PhD in Applied Mathematics 2024/03 – 2024/12

School of Applied Mathematics, Getulio Vargas Foundation, Brazil. Advisor: [Prof. Diego Mesquita](#).

- Thesis title: Streaming, Distributed, and Asynchronous Amortized Inference.
- Fast-tracked graduation due to recognized academic excellence and scientific productivity.

BSc in Data Science 2020/03 – 2023/12

School of Applied Mathematics, Getulio Vargas Foundation, Brazil

- GPA: 9.9/10.0. 1st in class and 1st in the entrance exam. Received a fully-funded scholarship.

Employment

Mohamed bin Zayed University of Artificial Intelligence *Abu Dhabi, UAE*

- Postdoctoral Associate under supervision of [Salem Lahlou](#). 2025/10 – ongoing
 - I am working on probabilistic machine learning and Bayesian inference.

Rei do Pitaco (fastest-growing sports betting startup in Brazil) *Santa Catarina, Brazil*

- Data Scientist & Engineer. 2025/01 – 2025/11
 - I spearheaded the development of the company’s bet builder, allowing customers to create customized bets. I designed, implemented, and maintained both the algorithm and the web API responsible for the real-time odds calculations.
 - Handling up to 1.2 million requests per day, the service accounted for 50% of the company’s sportbook revenue.
 - Developed ETL pipelines with dbt and Airflow for near-real-time data processing and service monitoring.

Aalto University *Espoo, Finland*

- Visiting scholar under the supervision of [Samuel Kaski](#) and [Vikas Garg](#). 2024/07 – 2024/10

Proffer (Price Monitoring & Optimization startup) *Rio de Janeiro, Brazil*

- Software Engineer. 2024/04 – 2025/09
 - I led the adoption of Infrastructure as Code (IaC) with Terraform as a standard for the company’s web scraping services.
 - I assisted in the migration of the company’s infrastructure between cloud providers (AWS and Azure).

Rei do Pitaco (largest fantasy sports company in Brazil) *Rio de Janeiro, Brazil*

- Data Science intern. 2023/01 – 2023/07
 - I designed predictive models to define the opening lines of bets on the outcomes of sport events (bookmaking).
 - I deployed and upheld the created models within applications serving thousands of concurrent users.

Selected Honors & Awards

Award for Academic Excellence, Brazilian Society of Applied and Computational Mathematics. 2023

First place, School of Applied Mathematics entrance exam. 2020

William Glenn Whitley Prize for achieving the highest score on the State Mathematical Olympiad. 2019

Top score in the country, Brazilian Mathematical Olympiad of Public Schools. 2018-2019

Selected Publications

- Path-dependent Discrete Amortized Inference** ICML 2026 (oral)
 - [da Silva, T.](#), Whitammer, E., Lahlou, S.
- When do GFlowNets learn the right distribution?** ICLR 2025 (spotlight)
 - [da Silva, T.](#), Silva, E., Alves, R., Souza, A., Garg, V., Kaski, S., Mesquita, D.
- Generalization and Distributed Learning of GFlowNets** ICLR 2025
 - [da Silva, T.](#), Souza, A., Rivasplata, O., Garg, V., Kaski, S., Mesquita D.
- Streaming Bayes GFlowNets** NeurIPS 2024
 - [da Silva, T.](#), Souza, D., and Mesquita, D.
- On Divergence Measures for Training GFlowNets** NeurIPS 2024
 - [da Silva, T.](#), Silva, E., and Mesquita, D.

Languages

Portuguese (Native), English

Skills

Computer languages: Proficient with [Python](#) and [SQL](#). Competent with R and Stan and Rust. Familiar with C++.

Scientific computing frameworks: PyTorch, PyTorch Geometric, GPyTorch, NumPy, SciPy, Jax, sklearn, Pandas.

Technologies: Git, Linux, Docker, FastAPI, Datadog, Kubernetes, Argo, Tailscale, Kafka, Airflow, dbt.

References

Diego Mesquita	diego.mesquita@fgv.br
Amauri Souza	amauri.souza@aalto.fi